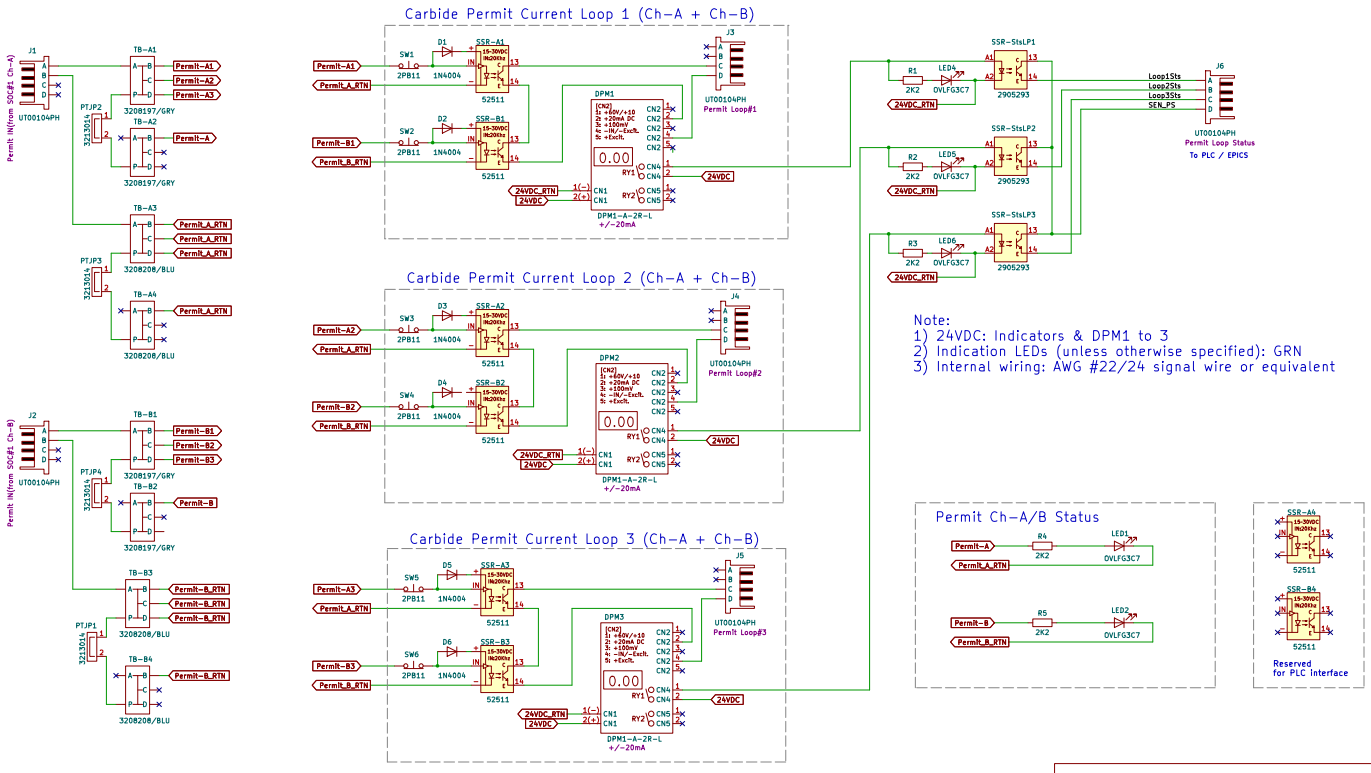
3x CARBIDE PERMIT INTERLOCK



DRFT: KICad v10	
CHKR:	
ENGR: J.Park	
SLAC	
Sheet: /3xCarbide Permit ILK/	
File: 3xCarbide Permit ILK.kicad_sch	
Title: LCLS-II BCS CARBIDE PERMIT INTERLOCK	
Size: B	Date: 2026-08-21
KICad E.D.A. 10.0.7	Rev: 1.03
	Id: 1/3

FRONT

CARBIDE PERMIT INTERFACE CHASSIS

Permit | Permit Removal Test PB | Carbide Permit Loop Status

0.00 | 0.00 | 0.00

SD-376-0XX-XX

REAR

SD-376-0XX-XX

(FROM SOC#1) A PERMIT J1 | B PERMIT J2

(TO CARBIDE) PERMIT LOOP1 J3 | PERMIT LOOP2 J4 | PERMIT LOOP3 J5

(TO PLC1) LOOP STATUS J6

F 3A

AC IN J7

For reference:
NOT TO SCALE

Carbide Permit Interface Cable Connection Example

(3x)



CARBIDE LASER (1-OF-3)

4Pin/F (UT0610-4SH)
A > N/C
B > N/C
C > XS13#21(5V:OUT)
D > XS13#9 (IN)
No conn. reserved
A/B Not use

AWG20
~3ft*
(*appropriate length)

UT0610-4SH
Souriau

BCS Permit Cable for Carbide XS13 Interface

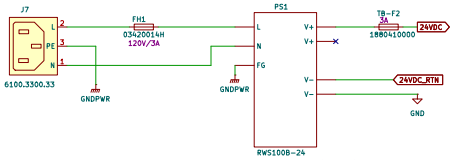
PARTIAL_LASER_STOP(X13.9-21): Current-loop (5V, 40 mA max.)

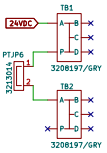
TOPL 5VDC 48VDC0.5A // Item No.2773760000
TERMSERIES-compact, Solid-state relay,
Rated control voltage: 5 V DC ±20 %
Rated switching voltage: 5..48 V DC,
Continuous current: 500 mA, PUSH IN

Rated control voltage: 5 V DC ±20 %
Nominal control current: 4.4 mA DC

Ref. CONN of the AOM chassis







Related accessories

Acc.EDP1
3208375
Conn.End Plate

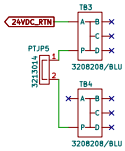
Acc.PLUG
A
B
C
D
UT0610SH

Acc.PIN1
RM16M23K

Acc.Socket
RC16M23D2B

Acc.Bcksh1
UT010AC





Key SSR
-Switch off characteristics

SSR(20KhZ).52511
Switch off delay max. 18 µs

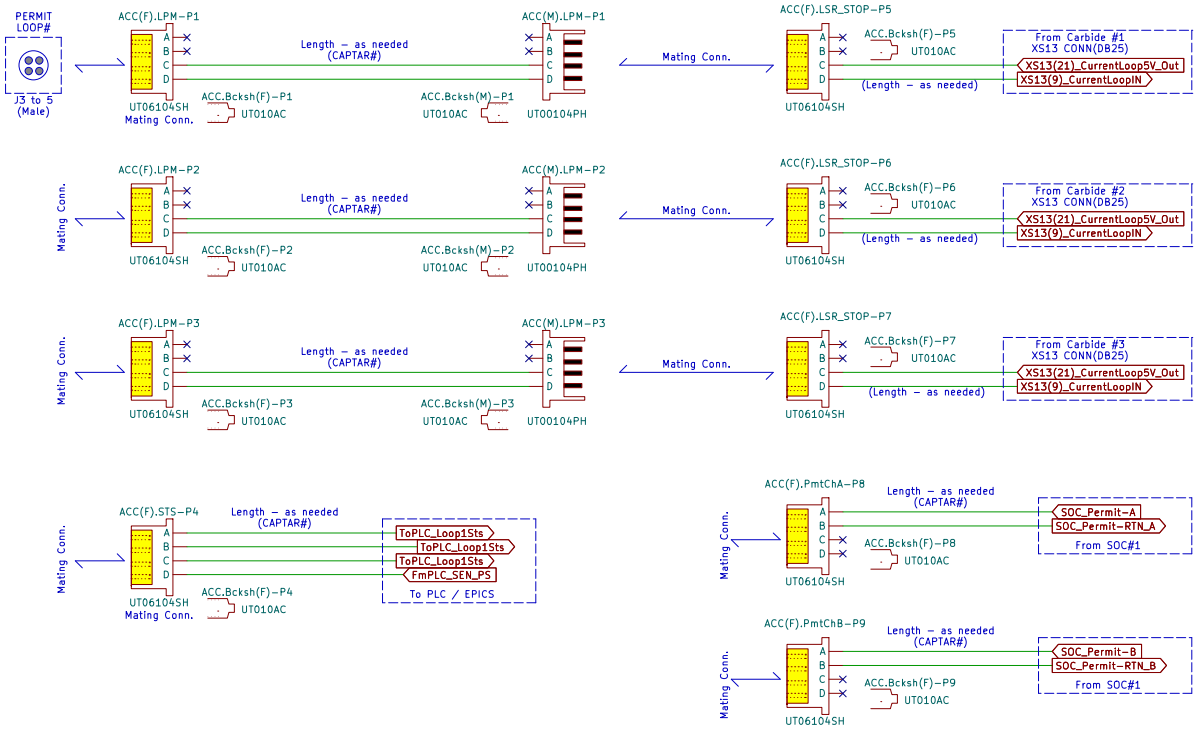


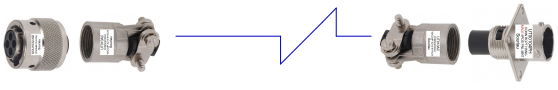
SSR.2905293
Typical turn-off time 300 µs



DRFT: KICad v10		
CHKR:		
ENGR: J.Park		
SLAC		
Sheet: /PS_CONN/		
File: conn.kicad_sch		
Title: LCLS-II BCS CARBIDE PERMIT INTERLOCK		
Size: B	Date: 2026-08-21	Rev: 1.03
KICad E.D.A. 10.0.7		Id: 2/3

Carbide Permit Interface Cable Assembly Drawing





Inline connector assembly

CHKR:	
ENGR: J.Park	
SLAC	
Sheet: /Interface Cable/	
File: interface_cable.kicad_sch	
Title: LCLS-II BCS CARBIDE PERMIT INTERLOCK	
Size: A4	Date: 2026-08-21
KiCad E.D.A. 10.0.7	**Rev: 1.04**
	Id: 3/3